



Vel Tech
Rangarajan Dr. Sagunthala
R&D Institute of Science and Technology
(Deemed to be University Estd. u/s 3 of UGC Act, 1956)



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WEEKLY LESSON PLAN

Department: Computer Science and Engineering

Year / Semester: 2025-26 / Winter

Course Code / Course Name: 10212CS292 / Applied Programming Skills

Unit I: Linear Data Structures for Competitive Input Processing				
Course Outcome: Use linear data structure techniques to process competitive programming inputs efficiently.				
Week	Session	Topics to be Covered	Problems to Practice (Hands-On)	Platform
Week 1	Session 1	Course Orientation Time Complexity Analysis Two Pointers, Sliding Window	Task 1: Squares of a Sorted Array Task 2: Contains Duplicate II	Leet Code
	Session 2	Prefix Sum Difference Array	Task 3: Running Sum of 1d Array Task 4: Sum of Absolute Differences in a Sorted Array	
	Session 3	Binary Search Sorting	Task 5: Binary Search Task 6: Sort Colors	
	Session 4	Greedy pairing Hashing for frequency/count	Task 7: Best Time to Buy and Sell Stock Task 8: First Unique Character in a String	
	Session 5	In-place array manipulation	Task 9: Move Zeroes Task 10: Reverse String	



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Week 2	Session 6	Linked list fast/slow pointer	Task 11: Middle of the Linked List Task 12: Palindrome Linked List	Leet Code
	Session 7	Merge two lists	Task 13: Merge Two Sorted Lists Task 14: Merge k Sorted Lists	
	Session 8	Reverse nodes in k-group	Task 15: Reverse Linked List Task 16: Reverse Nodes in k-Group	
	Session 9	Detect/remove cycle	Task 17: Linked List Cycle II	
	Session 10	Intersection of linked lists	Task 18: Intersection of Two Linked Lists	



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Unit II: Stack and Queue Based Competitive Problem Modeling Course Outcome: Demonstrate stack and queue operations to solve expression processing, scheduling, and data-handling problems.				
Week	Session	Topics to be Covered	Problems to Practice (Hands-On)	Platform
Week 3	Session 11	Monotonic Stack - Next Greater Element pattern	Task 19: Next Greater Element I Task 20: Daily Temperatures	Leet Code
	Session 12	Stock Span pattern	Task 21: Online Stock Span Task 22: Final Prices With a Special Discount in a Shop	
	Session 13	Valid parentheses variants	Task 23: Valid Parentheses Task 24: Minimum Remove to Make Valid Parentheses	
	Session 14	Min Stack design	Task 25: Min Stack	
	Session 15	Stack simulation problems	Task 26: Asteroid Collision Task 27: Validate Stack Sequences	

Week	Session	Topics to be Covered	Problems to Practice (Hands-On)	Platform
Week 4	Session 16	Queue using stacks	Task 28: Implement Queue using Stacks	Leet Code
	Session 17	Sliding window maximum using deque	Task 29: Sliding Window Maximum Task 30: Longest Continuous Subarray With Absolute Diff Less Than or Equal to Limit	
	Session 18	BFS-level processing with queue	Task 31: Binary Tree Level Order Traversal Task 32: Binary Tree Right Side View	
	Session 19	Circular queue simulation	Task 33: Design Circular Queue Task 34: Design Circular Deque	
	Session 20	Task scheduling with queue	Task 35: Task Scheduler Task 36: Number of Recent Calls	